



Food and Agriculture
Organization of the
United Nations



WEBINARS SERIES ON RAINWATER HARVESTING

5th October 2022

Module n°7:

**Water buffering for climate management
and biodiversity enhancement**



3R Land and Water Management **Plus**

The progressing approach to better water and land management

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Introduction to 3R Land and Water Management

Water harvesting at landscape level

- Recharge, retention, reuse

Key messages

Landscape level water harvesting

Work intensely, not scattered

Extend the 'chain of uses'

Managing the water buffer

Extend the 'chain of uses'

Transforming landscapes, transforming
lives



Introduction to 3R Land and Water Management

There are many possible (technical) solutions to achieve recharge, retention, and reuse.

Depending on:

- Landscape, slope
- Geology and soils
- Rainfall pattern
- Economy, land use

Tools for selection:

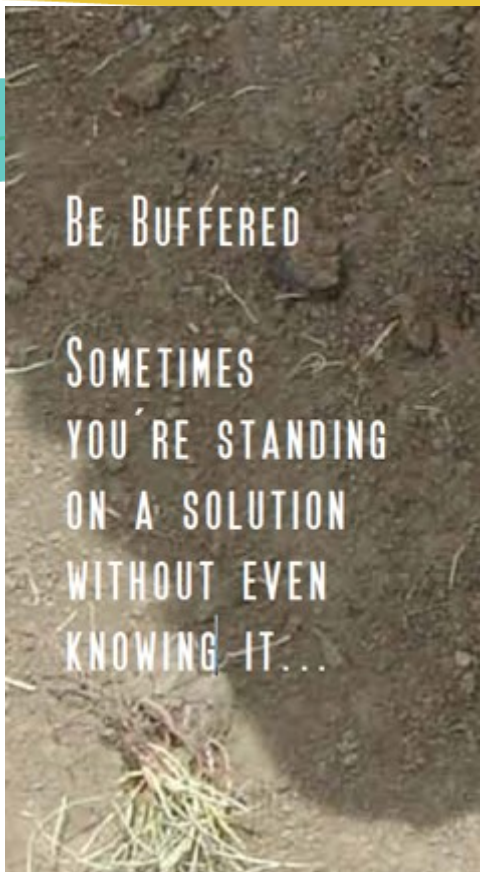
<https://sahel.acaciadata.com/#>

Reversal can be done fast!

**Treating a landscape in one year; Example
from Tigray, Ethiopia**

Landscape restoration and water harvesting
practices





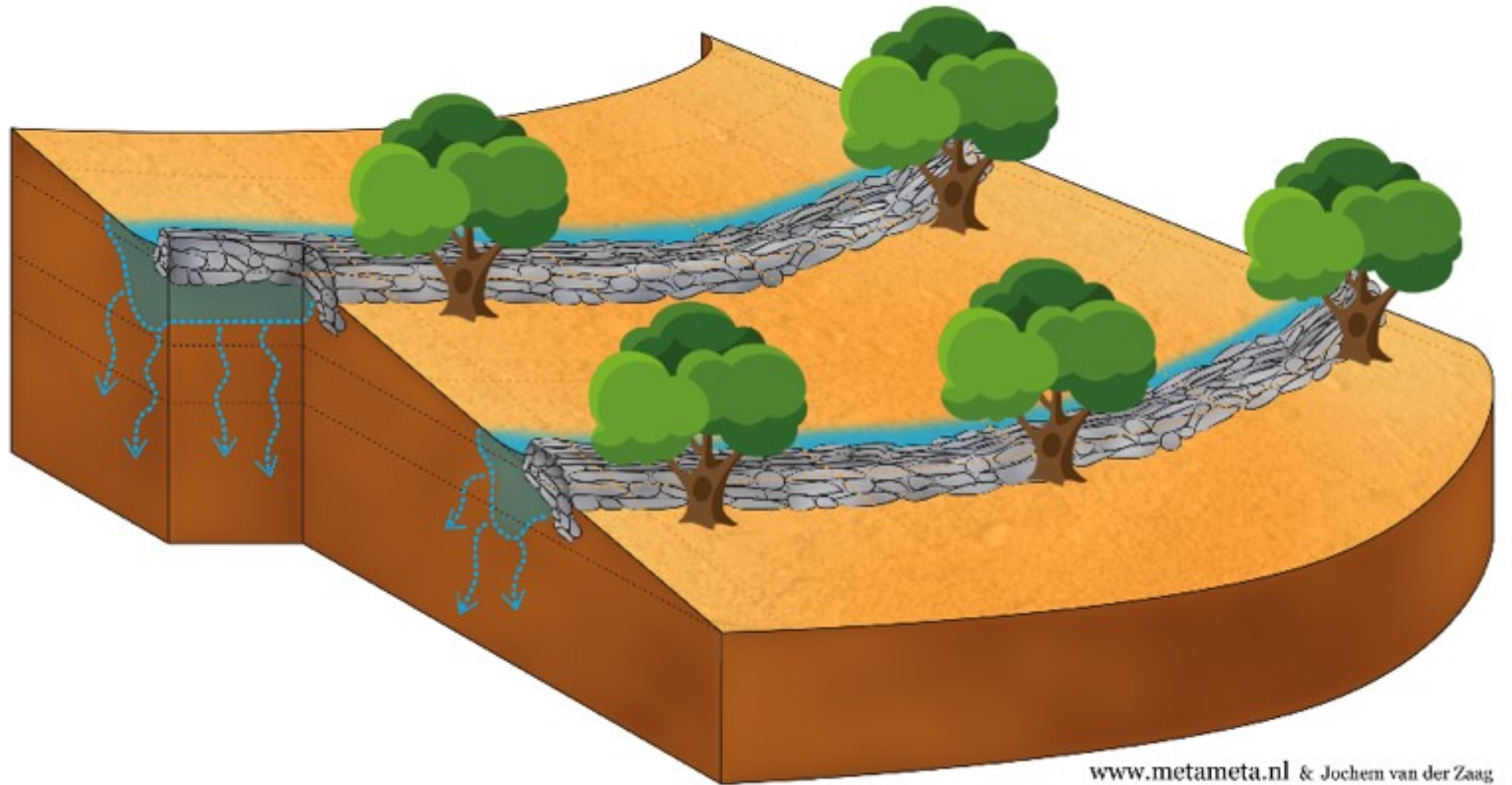
Guidelines to Good Practice

<https://www.wocat.net/library/media/25/>

Table 4: Overview of techniques described in this book

Technique			Geographic Suitability*			Function						New or existing
#	Main category		Upper	Middle	Lower	Soil conservation	Recharge	Retention	Reuse	Micro-climate	Water quality	
1	A	Contour bunds	•	•		•	•	•				Existing
2	A	Grass strips	•			•		•			•	Existing
3	A	Gully plugging	•			•	•	•				Existing
4	A	Bench terraces	•			•		•				Existing
5	A	Stone bunds	•	•		•	•	•				Existing
6	A	Trapezoidal bunds		•	•		•	•				Existing
7	A	Tied ridges	•			•		•				New
8	A	Demi lunes		•		•		•				Existing
9	A	Tal ya trays			•							Existing
10	A	Double dug beds			•			•		•		New
11	A	Composting	•	•	•			•		•		New
12	A	Bio-char	•	•	•			•				New
13	A	Organic mulching	•	•	•			•		•		Existing
14	A	Plastic mulching		•	•			•		•		Existing
15	A	Making use of invertebrates		•	•		•					New
16	A	Planting pits		•	•	•	•	•				Existing
17	B	Percolation ponds and contour trenches	•	•		•	•					New
18	B	Tube recharge	•				•					New
19	B	Subsurface dams	•	•	•		•	•	•			Existing
20	B	Sand dams		•	•		•	•	•			Existing
21	B	Sand dune water infiltration			•		•		•			New
22	C	Harvesting water from roads		•	•		•	•	•			Existing

- Maintains the runoff
- Increases soil moisture levels
- Increases groundwater recharge
- Buffers local temperatures



www.metameta.nl & Jochem van der Zaag

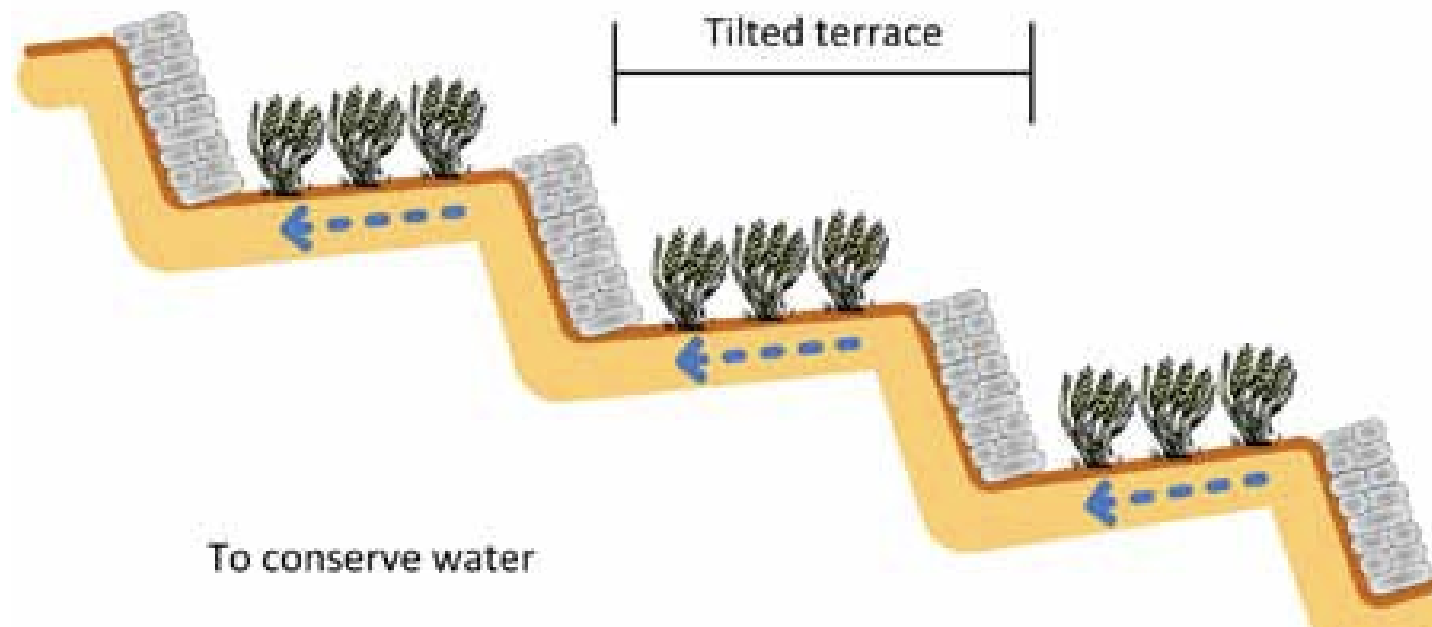
Eyebrow Terraces and Stone Bunds



Source: Justdigg.it



Soil bunds



Terracing



Rock catchment for water harvesting



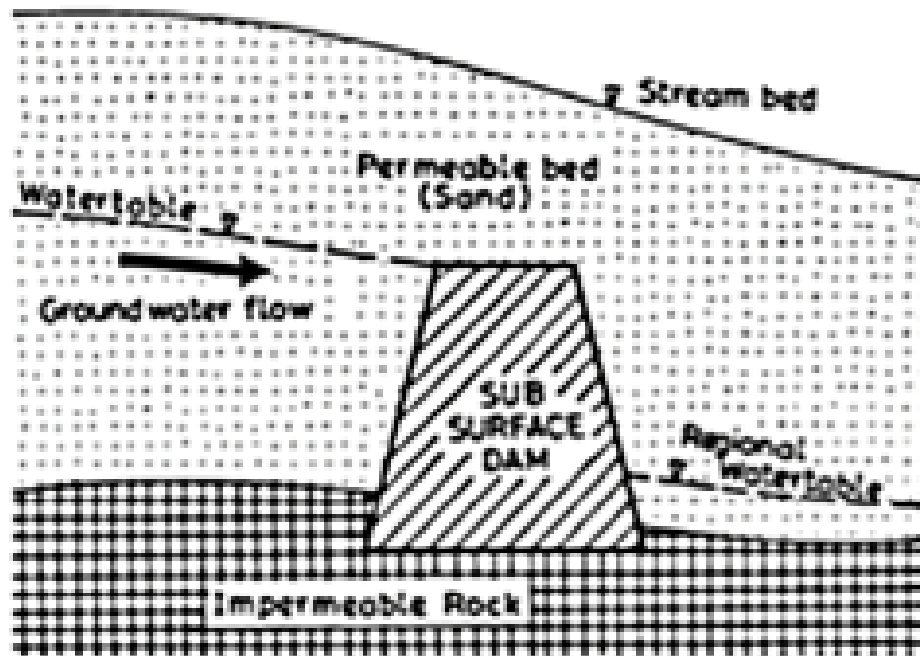
Using roads for groundwater recharge



- During peak discharge season, water is impounded behind the weir
- In the dry season, the water is released, and sediment flushed



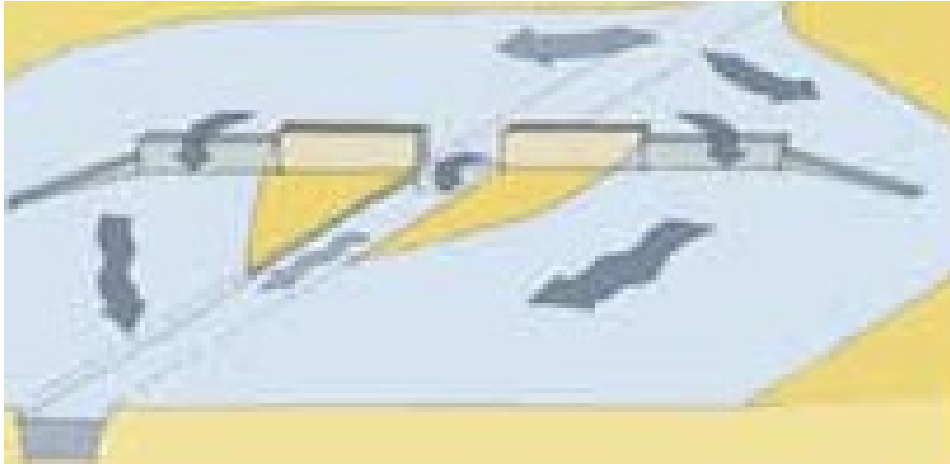
Weir to feed the aquifer, Maharashtra - India



Subsurface dam



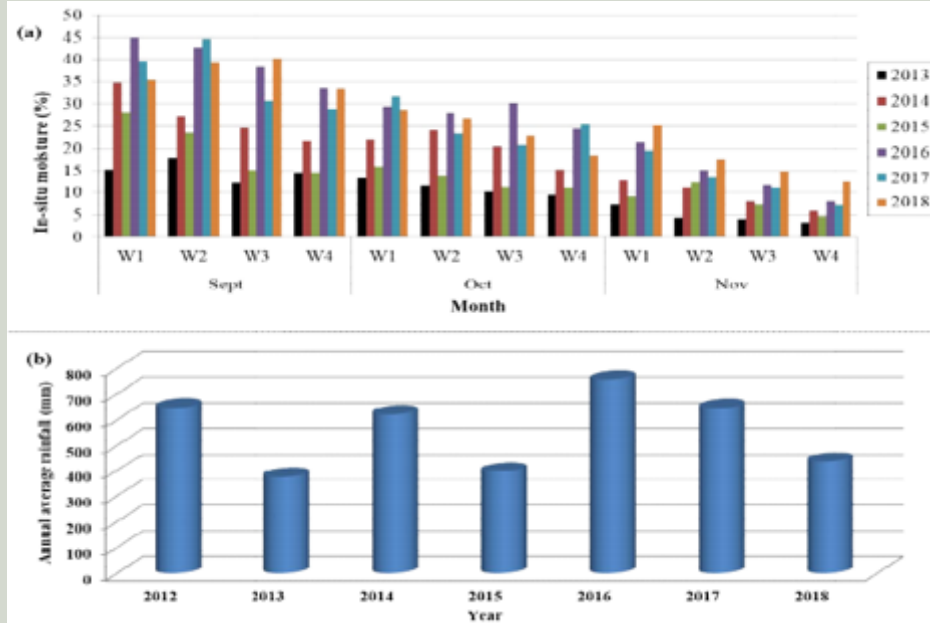
Road drift aka sand dam



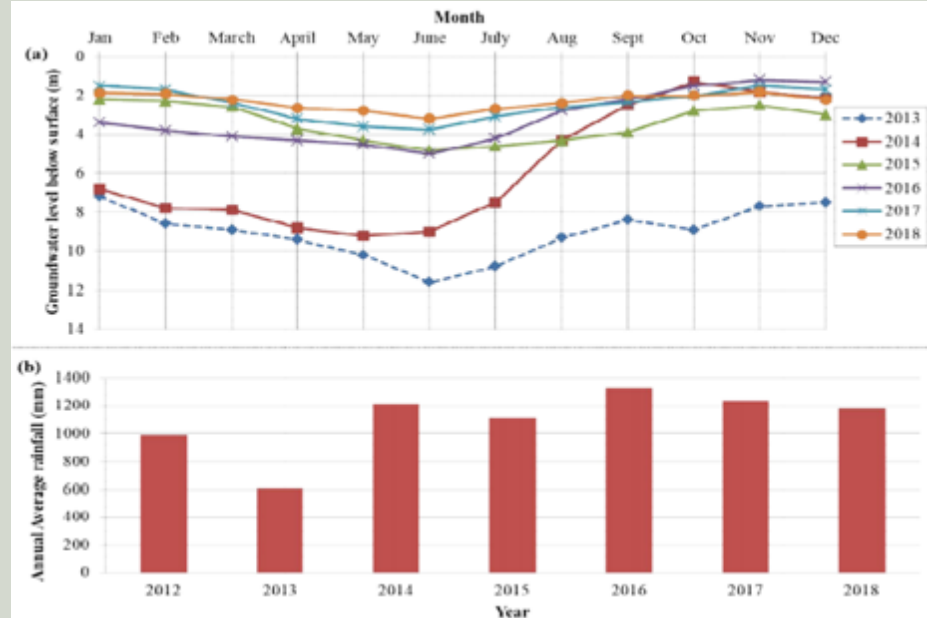
Flood water spreading weir

Monitoring the impact of Green Roads in Ethiopia

Kifle, W., Berhane, G., Taye, A., Kebede, M., & Marta, A. P. (2019). Practices and Hydrological Effects of Road Water Harvesting in Northern Ethiopia: Towards Design of Multi-Functional Infrastructures. *Momona Ethiopian Journal of Science*, 11(2), 159-186



(a) In-situ moisture distribution in soils (before and after the construction of structures that divert runoff from culverts into farmlands along the Mekelle road (Kihen), Tigray, Ethiopia. Construction of the diversion structures was done on May-June 2014. Monitoring was done for the period September years 2013 to 2018. (W1= Week one; W2=Week two; W3=Week three and W4=Week four). (b) Rainfall distribution for the year 2012 to 2018. (ENMSA, 2018).



a) Groundwater fluctuation in Selekleka area, Tigray, Ethiopia (at downstream of a check-dam which was constructed in the period January is designed to store improved at downstream of the box culvert b) Rainfall distribution for the year 2012 to 2018 (ENMSA, 2018)



Key features

- Community driven
- Every place has own options and solutions
- Beyond water harvesting
 - Land use measures (area closure, controlled grazing)
 - Agronomic and regreening measures

Go beyond: 3R Land and Water Management **Plus**

Diversify 3R plus

- Managed regreening
- Management of soil biota
- Improving the local climate



Managed regreening

- Restoring common forests and hedges in 3R landscapes has major benefits for nearby communities:
 - Bring back rainfall
 - Recharge groundwater
 - Green biomass production > basis for biofertilizer
 - Improved grazing areas for livestock



Villagers checking upon the forest in the evening in Madhya Pradesh, India (Photo taken by Rajnish Mishra)

Managed regreening

- Restoring forests and hedges in 3R landscapes has major benefits for nearby communities :
- Depending on the type of trees or greenery:
 - Fruit or seeds production from trees/shrubs
 - Timber, fuelwood
 - Fodder production
 - Bee forage
 - Compost production
 - Provision of organic fertilizer inputs
 - Wind breaks



Hedging opportunities
<https://thewaterchannel.tv/thewaterblog/hedging-opportunities/>

Managed regreening

- Special case of pastoral lowlands
- Controlled holistic grazing
 - More infiltration (hoof action)
 - More fertility (droppings)
 - Better biodiversity (all species)

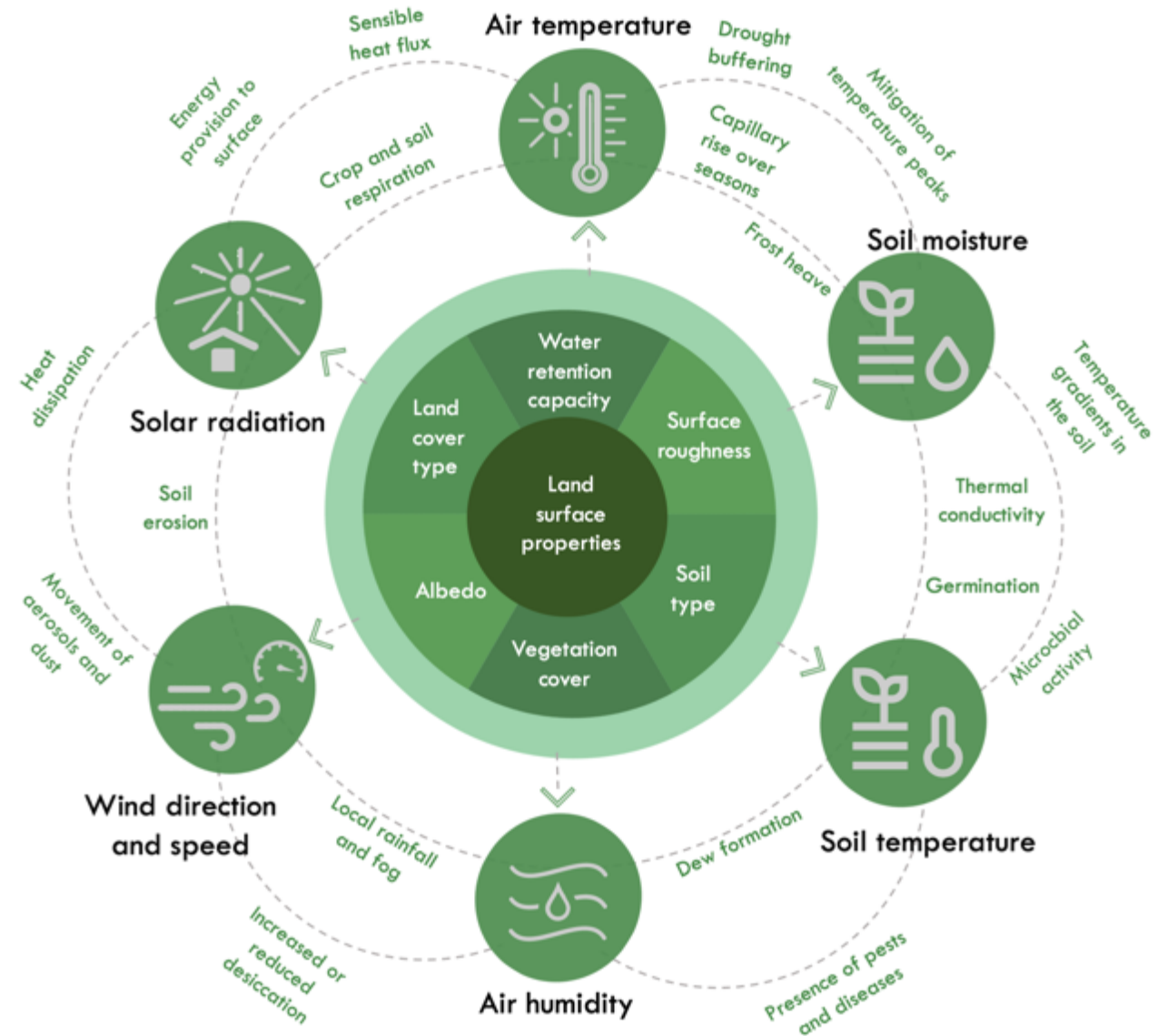


Management of soil biota

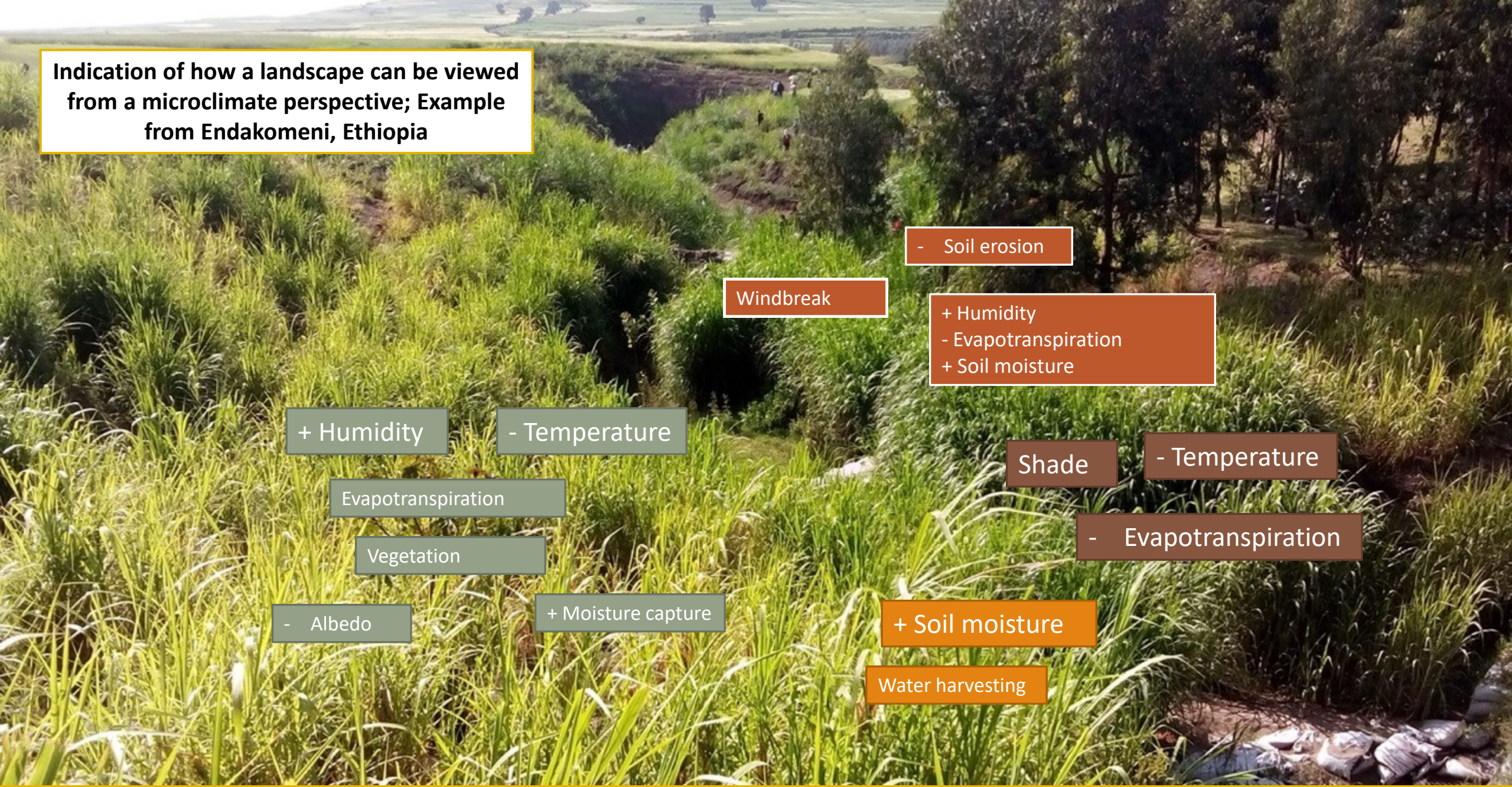


Floodwater spreading and sowbug, Iran

Management of local climates



Indication of how a landscape can be viewed from a microclimate perspective; Example from Endakomeni, Ethiopia



- Soil erosion

Windbreak

+ Humidity
- Evapotranspiration
+ Soil moisture

+ Humidity

- Temperature

Evapotranspiration

Vegetation

- Albedo

+ Moisture capture

Shade

- Temperature

- Evapotranspiration

+ Soil moisture

Water harvesting

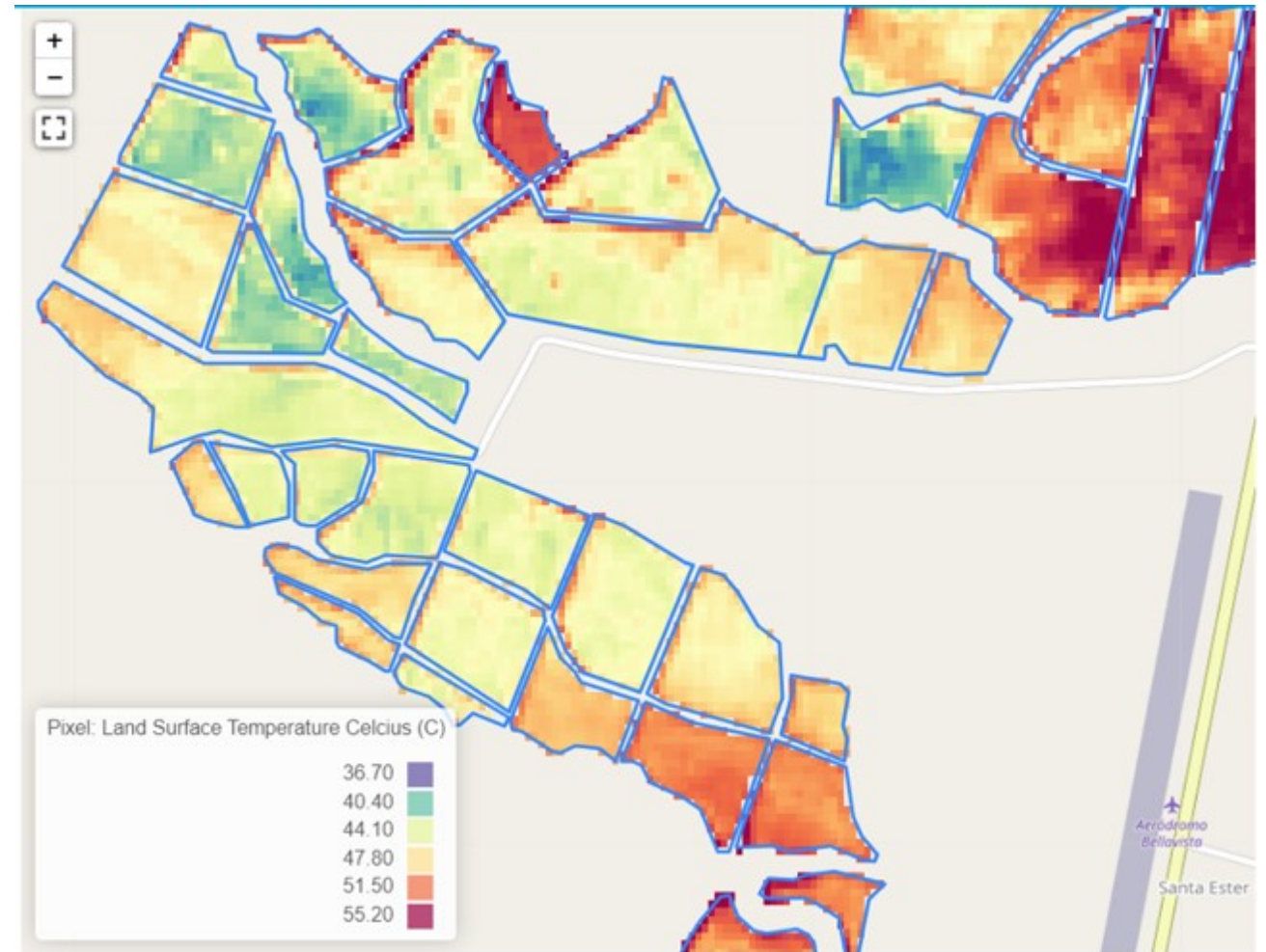
Local climate management

- The high variability of local land surface properties results in a mosaic of different micro- and macro climates spread over landscapes



Local climate management

- Even on farm scale there can be significant differences in microclimate



Thermal image of an orchard in Chili with a detail of 10-meter x 10- meter acquired on 26th of December 2021 -
Source: Irriwatch



Local climate management

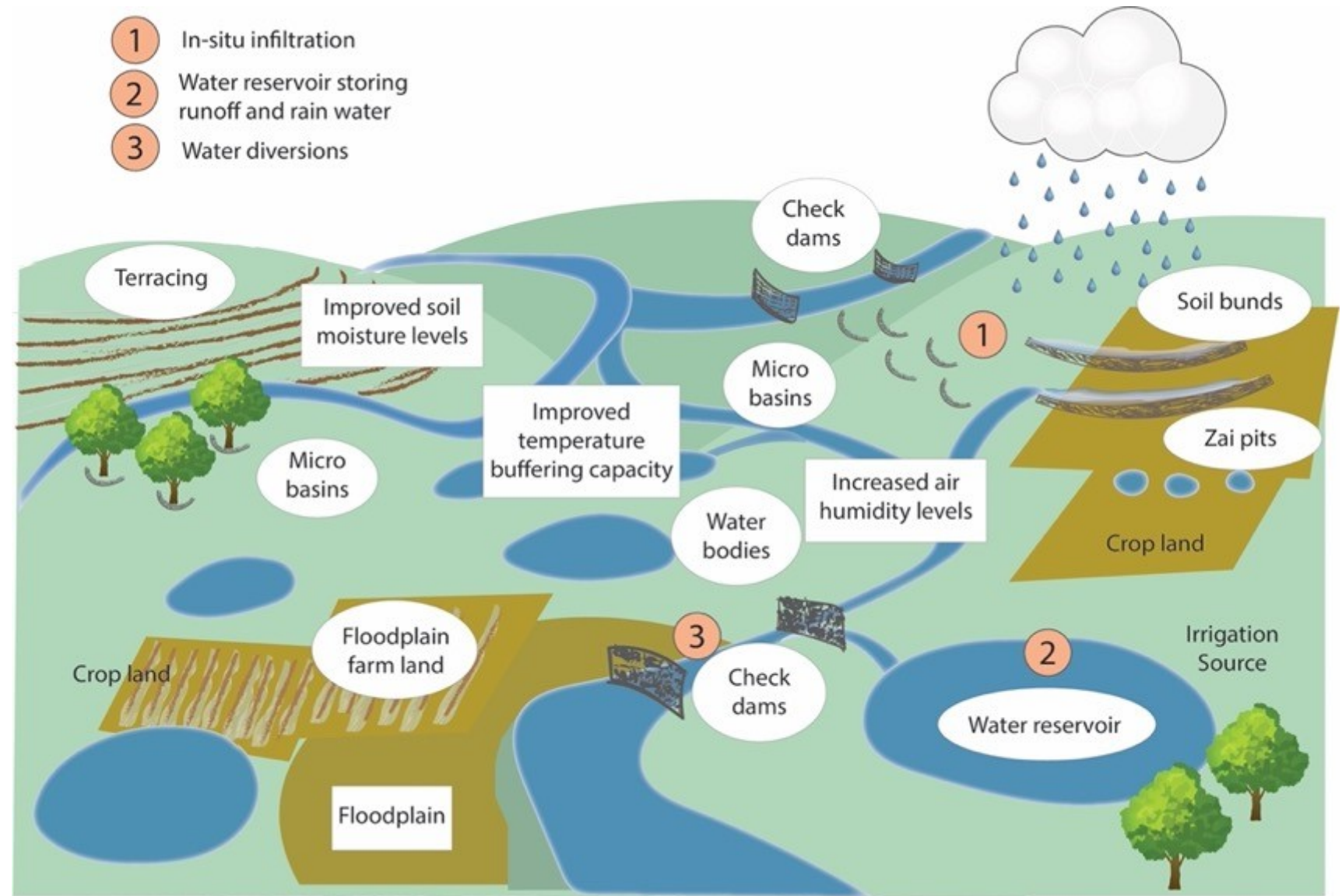
- The fact that different landscape characteristics result in different local climates is good news:
- We can intervene in the landscape, and through that, improve the local climate!

Local climate management

- ‘Third Way’ to deal with Climate Change, besides adaptation and mitigation
 - Temperature reductions (1.5-2 degrees) in the global warming bandwidth
 - Can be managed locally
 - Can be monitored
 - Large co-benefits
 - Lower temperatures and increased buffering capacity to climatic extremes result in more suitable agricultural conditions
 - Less crop damage
 - Higher and more diverse crop production

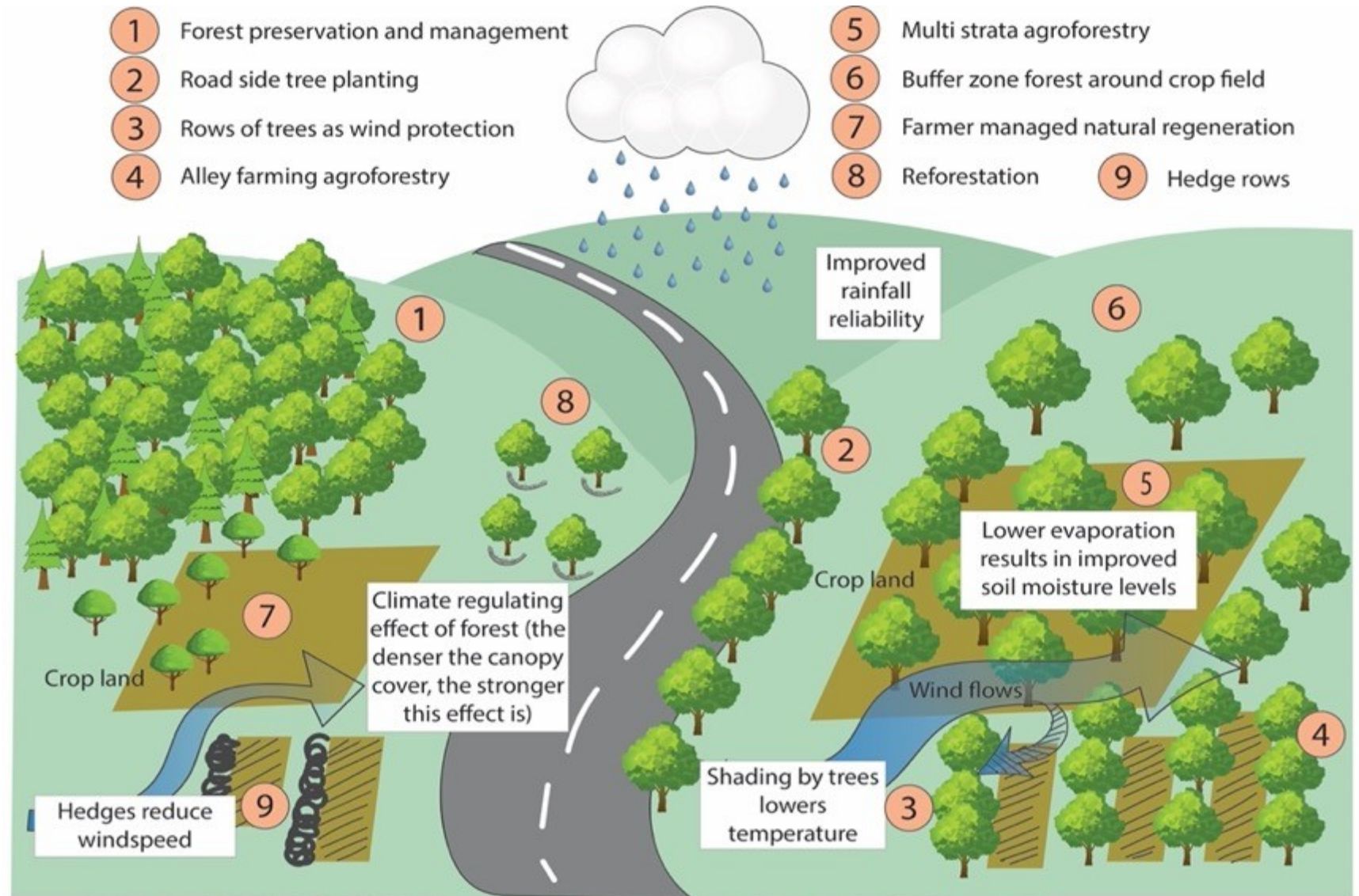
Local climate management

- Different land and water management techniques can be combined to create most optimal local climate



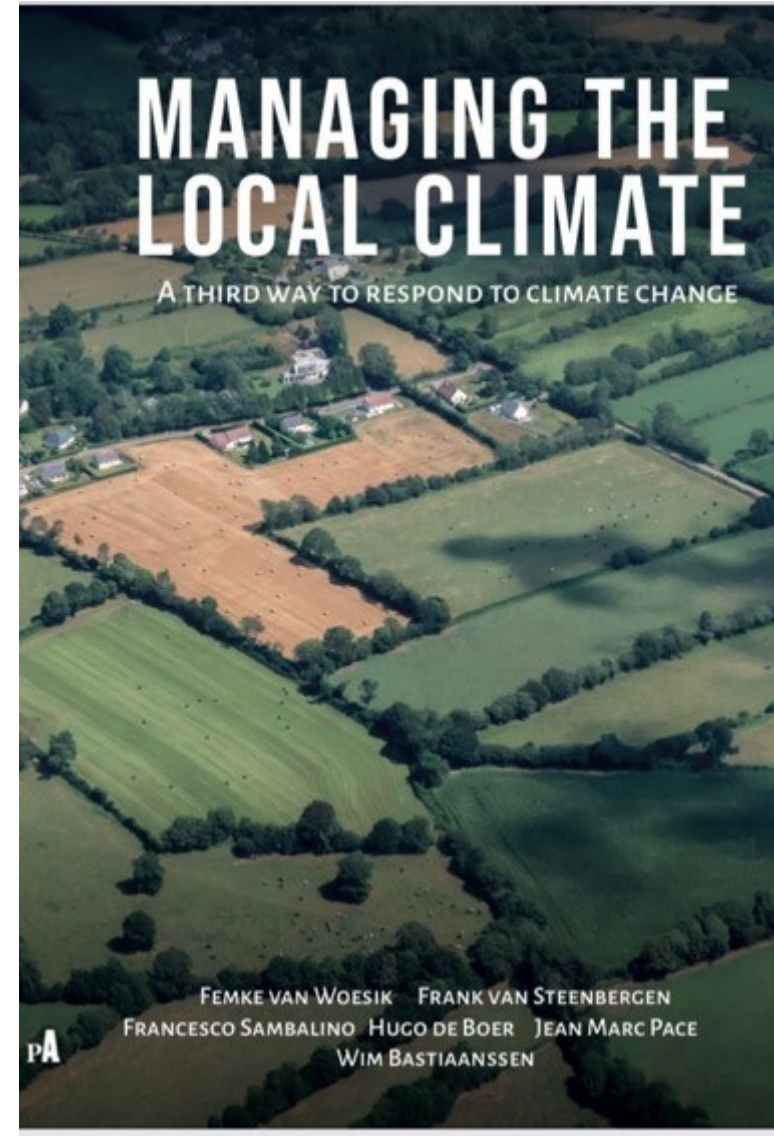
Local climate management

- Different land and water management techniques can be combined to create most optimal local climate



Local climate Management Publication

Van Woesik, F., Van Steenberg, F., Sambalino, F., De Boer, H., Pace, J. M., & Bastiaanssen, W. (2023). Managing the Local Climate – A third way to respond to climate change. Practical Action Publishing





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Thank you for
your attention